```
import pandas as pd
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files gender_classification.csv

gender_classification.csv(text/csv) - 128316 bytes, last modified: 4/29/2026 - 100% done

```
Saving gender_classification.csv to gender_classification.csv
```

```
df=pd.read_csv("gender_classification.csv")
```

```
df
```

	long_hair	forehead_width_cm	forehead_height_cm	nose_wide	nose_long	li
0	1	11.8	6.1	1	0	
1	0	14.0	5.4	0	0	
2	0	11.8	6.3	1	1	
3	0	14.4	6.1	0	1	
4	1	13.5	5.9	0	0	
...	...	...	...	...	...	
4996	1	13.6	5.1	0	0	
4997	1	11.9	5.4	0	0	
4998	1	12.9	5.7	0	0	
4999	1	13.2	6.2	0	0	
5000	1	15.4	5.4	1	1	

5001 rows × 8 columns

Next steps: Generate code with df | New interactive sheet

```
print("Missing values:\n", df.isnull().sum())
```

```
Missing values:
 long_hair                  0
forehead_width_cm           0
forehead_height_cm          0
nose_wide                   0
nose_long                   0
```

```
lips_thin                    0
distance_nose_to_lip_long    0
gender                       0
dtype: int64
```

```
print("\nDuplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 1768
```

```
df = df.drop_duplicates()
```

```
print("\nDuplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

```
from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()
df["gender"] = le.fit_transform(df["gender"])
```

```
/tmp/ipykernel_1957/3094290749.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stab
  df["gender"] = le.fit_transform(df["gender"])
```

```
X = df.drop("gender", axis=1)
y = df["gender"]
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random
```

```
from sklearn.linear_model import LogisticRegression
```

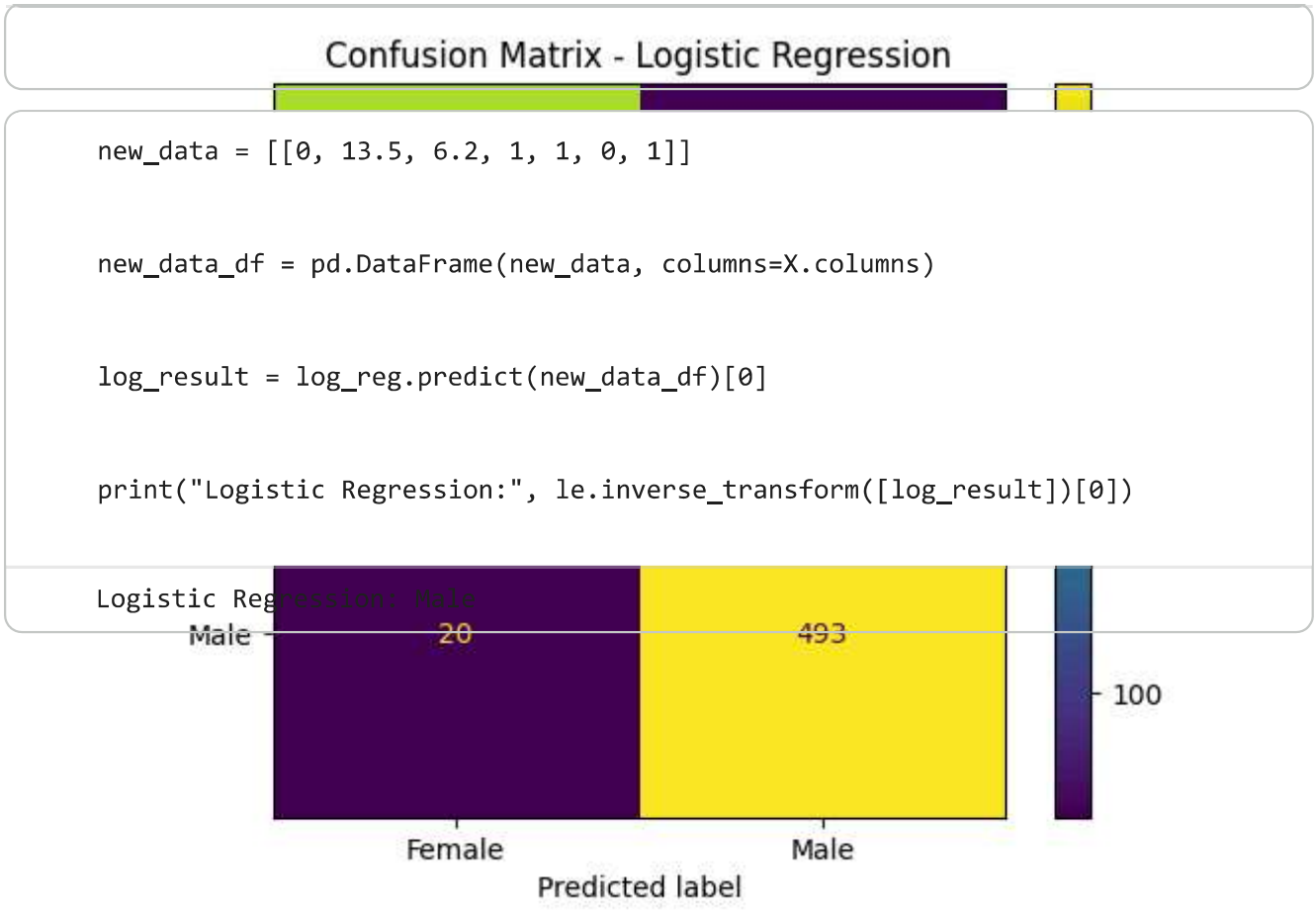
```
log_reg = LogisticRegression()
log_reg.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

```
from sklearn.metrics import accuracy_score, classification_report

y_pred_log = log_reg.predict(X_test)
print("\nLogistic Regression Results")
print("Accuracy:", accuracy_score(y_test, y_pred_log))
print(classification_report(y_test, y_pred_log, target_names=["Female","Male"])
```

```
Logistic Regression Results
Accuracy: 0.9556701030927836
              precision    recall  f1-score   support

      Female       0.96      0.95      0.95       457
        Male       0.96      0.96      0.96       513

    accuracy                           0.96       970
   macro avg       0.96      0.96      0.96       970
weighted avg       0.96      0.96      0.96       970
```

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred_log)

disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Female", "Ma
disp.plot()
plt.title("Confusion Matrix - Logistic Regression")
plt.show()
```




```
new_data = [[0, 13.5, 6.2, 1, 1, 0, 1]]

new_data_df = pd.DataFrame(new_data, columns=X.columns)

log_result = log_reg.predict(new_data_df)[0]

print("Logistic Regression:", le.inverse_transform([log_result])[0])
```

```
Logistic Regression: Male
```